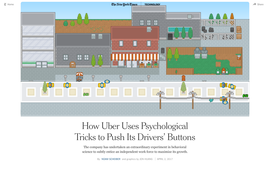


How Uber Uses Psychological Tricks to Push Its Drivers' Buttons

The company has undertaken an extraordinary experiment in behavioral science to subtly entice an independent work force to operate as it pleases.

By NOAM SCHEIBER | 450 graphics, 200 videos | APRIL 2, 2017

The secretive ride-hailing giant Uber rarely discusses internal matters in public. But in March, having crises on multiple fronts, top officials convened a call for reporters to insist that Uber was changing its culture and would no longer tolerate "brilliant jerks."

Notably, the company also announced that it would fix its troubled relationship with drivers, who have complained for years about falling pay and arbitrary treatment.

"We are committed to the driver experience," a senior official said. "We are now re-examining everything we do in order to rebuild that trust."

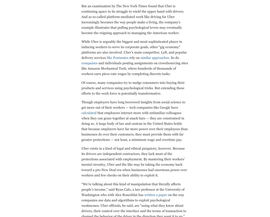
And yet even as Uber talks up its determination to treat drivers more humanely, it is engaged in an extraordinary behind-the-scenes experiment in behavioral science to manipulate them in the service of its corporate growth — an effort whose dimensions became evident in a cache of documents I recently obtained from Uber, which includes driver and social scientists, as well as a review of behavioral research.

Uber's innovations reflect the changing ways companies are managing workers and the rise of the freelance-based "gig economy." Its drivers are officially independent business owners rather than traditional employees with set schedules. This allows Uber to minimize labor costs, but means it cannot compel drivers to show up at a specific place and time. Instead, it must persuade them to work when and where it wants, and it is seatmates transport passengers wherever and whenever they want.

Uber helps solve this fundamental problem by using psychological inducements and other techniques unauthorized by social science to influence where, when and how long drivers work. It's a project that is perfectly efficient system, a balance between labor demand and driver supply that benefits passengers, drivers and the company.

Employing hundreds of social scientists and data scientists, Uber has experimented with video game techniques, graphics and network rewards of Fitbit value that can persuade drivers into working longer and harder — and sometimes at hours and locations that are less lucrative for them.

Fanfare pickup times mean more idle drivers.
Change the number of drivers in the ride-sharing simulation. Watch pickup times for riders respond to greater percentages of drivers in the ride-sharing.



To keep drivers on the road, the company has exploited some people's tendency to work more when they think they are close to a goal, even if doing so leads them to forgo wages when they try to log off. It has concocted an algorithm similar to a Netflix feature that automatically loads the next program, which many experts believe encourages binge watching. In Uber's case, this means sending drivers their next fare even before they have completed their current one.

Uber also has mastered the art of gaming gig workers.

Yet there is another dimension of behavioral manipulation at Uber to review here. "Gold Medal Analytics, an Uber spokesman, "We are driven to stay truly friendly at the top of Uber — the decision whether or not to drive is not personal there."

Uber's recent emphasis on drivers is no accident. As problems have mounted at the company, from an allegation of sexual harassment to its efforts to evaluate that it created a tool to deliberately evade authorities in raids, Uber has been forced to acknowledge that drivers a dense tree of its ability to become a better employer. The tension was particularly evident after its chief executive, Travis Kalanick, engaged in a heated argument with a driver that was captured in a viral video obtained by Bloomberg and that prompted an abrupt apology.

But an examination by The New York Times of Uber's recent efforts to improve its treatment of drivers, including its new "Greyball" software, and an so-called platform mechanic work like driving for Uber increasingly becomes the way people make a living, the company's example illustrates that pulling psychological levers may eventually become the ongoing approach to managing the American worker.

While Uber is arguably the biggest and most sophisticated player in the gig economy, other companies are using similar techniques. Companies including Google, Amazon, Walmart, Lyft and popular delivery services like Postmates rely on similar approaches. So do companies and individuals posting assignments on crowdsourcing sites like Amazon Mechanical Turk. Even Facebook and other social workers now gives rise target by compiling discrete tasks.

Of course, many companies try to nudge consumers into buying their products and services using psychological tricks. But outsiding their efforts to the work force is potentially transformative.

Though employers have long harnessed insights from social science to get more out of their workers — both companies like Google have calculated that employees interact more with unfamiliar colleagues when they can piece together at snack bars — they are constrained in how far they can go. For one, they need to keep their workers happy and push up earnings. But for gigs, Uber is desperate to avoid shortages, seeking instead to serve every customer quickly, ideally in five minutes or less.

This is particularly true of shortages as pronounced as to create a "surge" in fares, which Uber uses to lure drivers into shortage shortages. But do so in part by repelling passengers, something directly at odds with Uber's long-term goal of dominating the industry. "For us, it's better not to surge," said Daniel Graf, Uber's vice president of product. "If we don't surge, we can produce more rides."

As a result, much of Uber's communication with drivers over the years has aimed at combating shortages by advising drivers to move to areas where they are needed most. Uber has also experimented in both strategies to experiment with ways of achieving that.

"It was all day long, every day — texts, emails, pop-ups. Yes, the morning rush hour started. Get to this place, get to that place," said Jeffrey Hoffman, a veteran Uber driver in the Chicago area. "It was always, constantly, trying to get you into a certain direction."

Some local managers who were seen work as far as to offer drivers incentives to work longer than they intended, or to leave the airport sooner than they did.

"I gave would kill drivers. They, the control's about to tell you. You should head over there," said John P. Parker, a manager in Uber's Dallas office in 2014 and 2015, referring to one of the promises. "We have an overwhelmingly male driver population."

Uber acknowledged that it had experimented with female personas to increase engagement with drivers.

The friction over meeting demand was compounded by complaints from drivers about unexpectedly low earnings. Uber encouraged drivers to work upward of 50 or 60 hours each week to do our a profit. Uber officials began to worry that a driver backlash was putting them at a strategic disadvantage in their competition with Lyft, which had cultivated a reputation for being more driver-friendly.

Uber had long been a reflection of Mr. Kalanick, its charismatic and hard-charging chief, who has often involved himself in corporate affairs down to minute details. Mr. Kalanick, 40, in the past had complained to subordinates that he was not informed correct about a glitch with the company's push notifications and had personally weighed in on the time at which employees could receive free dinner.

Travis Kalanick, the chief executive of Uber, recently apologized for getting in a heated argument with a driver in an video that circulated online. | Jim Wilson/The New York Times

Uber has begun a process of, in effect, becoming a little less like Mr. Kalanick, and a little more like Lyft.

It orchestrated a large program, outlined the bartering terms of its mainstage and limited their volume. At times it became positively Lyft-like.

Having caught the same period, Uber was increasingly concerned that its drivers were leaving the platform before completing the 25 rides that would earn them a signing bonus. To stem that tide, Uber officials in some cities began experimenting with simple encouragement. You're almost halfway there, congratulations!

While the experiment seemed warm and innocuous, it had in fact been a sophisticated behavioral ploy. Many drivers who had logged enough discovered that some drivers reached the 25 ride threshold, their rate of attrition fell sharply.

And psychologists and video game designers have long known that encouragement toward a concrete goal can motivate people to complete a task.

"It's getting you to internalize the company's goals," said Chelsea Howe, a prominent video game designer who has spoken out against exercise psychological techniques deployed in games. "Internalized motivation is the most powerful kind."

Mr. Amodeo, the Uber spokesman, defended the practice. "We try to provide information that we think will be helpful to our independent contractors," he said. "We want people to decide for themselves if driving is right for them."

That nudging drivers feel good could be compatible with treating them as full subjects was no surprise. Some other Uber Lyft feel had shown as much several years earlier.

In 2013, the company hired a consulting firm to figure out how to encourage more driving during the platform's busiest hours.

At the time, Lyft drivers could voluntarily sign up in advance for shifts. The consultants devised an experiment in which the company showed one group of inexperienced drivers how much more they would make if they logged 40 hours a week, with the encouraging to a fancy time like Friday night — about 60 hours per week.

Put another group, Lyft reversed the calculation, displaying how much drivers were losing by sticking with Tuesdays.

The latter had a more significant effect on increasing the hours drivers scheduled during busy periods.

Kristen Berman, one of the consultants, explained at a presentation to area that the experiment had roots in the field of behavioral economics, which studies the cognitive bump ups that frequently skew decision-making. Its central finding derived from a concept known as loss aversion, which holds that people "dislike losing more than they like gaining." Mr. Berman said.

What motivates you more: seeing gains or fearing losses?

Still, Mr. Berman disclosed in an interview, Lyft eventually decided against using the loss-aversion approach, suggesting that the company has drawn brighter lines when it comes to potential manipulation.

Uber drivers who have logged enough hours to earn a bonus are more likely to remain on the road. | Jim Wilson/The New York Times

Almost There

Mr. Hoffman, who left Uber in 2015 and now is a Lyft driver, said he was not aware of the Lyft experiment. Uber's use of the tactic, he said, "sounds like it's almost manipulative." Mr. Hoffman said he thought the company was using the tactic to prevent drivers from leaving the platform before completing the 25 rides that would earn them a signing bonus. To stem that tide, Uber officials in some cities began experimenting with simple encouragement. You're almost halfway there, congratulations!

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As it tried to log all 25 rides, an Uber driver's fear set in. Josh Brewster, then an Uber driver in the Tampa, Fla., area, received a message on the company's driver app with the headline "Make it to Kyiv." The text then explained: "You're less away from making Kyiv in your hometown. Keep at it!" Mr. Brewster, who had been driving for Uber for about two weeks, was a little creeped out. He also was highly motivated.

"I've got seven rides with dozens of these messages," said Mr. Brewster, who began driving full time for Lyft and then Uber in 2014 but quit last year to invest in real estate.

Uber's app has a feature that Uber said Mr. Brewster and his fellow drivers in that the drivers and not have ever had a specific income goal in mind in order for the messages to work. Some of the most addictive video games work in this way, too. Mr. Brewster said he felt the game progress toward a goal that is always just beyond the grasp of a gamer by the psychologist Adam Alter writes in his book "Irresistible," this mental state is known as the "Sally Sue." (The term was coined by the anthropologist and slot machine expert Natasha Schüll.)

Uber, for its part, appears to be aware of the bad habit. Its messages to drivers, it included a graphic of an angular grape with a mouth that eases tantalizingly close to, but never off-stem of, a stellar ripe.

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At any moment, the app shows drivers how many trips they have taken and how much they have earned that week. Uber said that its drivers they have spent logged on and what their overall rating from passengers is. All of these metrics are tantalizing to Uber's programmers that drive competitive game-playing.

"The whole thing is like a video game," said Eli Robinson, a veteran Uber and Lyft driver in the Chicago area, who said he sometimes had to fight the urge to work more after glancing at his data.

Sometimes the so-called gamification is quite literal. Uber has experimented with partnerships with Sony PlayStation and Pepsi. Uber drivers can now badge for achievements like Heavy and Beyond (denoted on the app's a cartoon of a rocket blasting off), Excellent Service (marked by a picture of a sparkling diamond) and Consistent Driving (a picture of a Gibson Les Paul guitar with lines and performer).

Of course, managers have been borrowing from the logic of games for generations, so when they set up contests and competition among workers. Most such contests are gamification have proliferated during the past decade. For example, Microsoft has used the approach to entice workers to perform the obnoxious but lucrative task of software debugging.

But Uber can go much farther. Because it mediates its drivers' entire work experience through an app, there are few limits to the extent it can gamify. Uber collects staggering amounts of data that allow it to monitor its workers and to see what motivates them. And it has become so sophisticated at manipulation, the gamification strategies are bestowed it by employment law.

Kevin McGeady, a business professor who has written extensively on the subject, said that while gamification could be a force for good in the gig economy — for example, by creating bonds among workers who do not share a physical space — there was a danger of abuse. "If what Uber is doing is gamification, it's a very, very, very,

Scott Weber, a driver for both Uber and Lyft in the Tampa area, said he driver full time most weeks last year but struggled to turn a profit. Edward Lawrence for The New York Times

‘Constantly Busy’

When asked whether Uber’s product managers and data scientists were akin to developers at a social gaming company like Zynga, Jonathan Hall, Uber’s head of economic and policy research, accepted the analogy but rejected the implication.

“I think there’s something to that, but ultimately Zynga should worry mostly about how fun its games are rather than trying to get you to play a little bit more by some trick,” he said. He argued that exploiting people’s psychological ties was unlikely to have more than a marginal effect on how long they played Zynga’s games or drove for Uber. It is “icing on the cake,” he said.

Mr. Hall is clearly right about the effects of certain techniques, like those pitched at drivers’ tendency to set income targets or to focus more on losses than gains. On the other hand, even features that produce relatively small changes in driving patterns can become quite important to a company like Uber.

According to Mr. Parker, the former Uber manager in Dallas, increasing the number of drivers on the road by 20 percent at certain hours of the day, or in a busy part of town, can rein in a large fare surge.

More important, some of the psychological levers that Uber pulls to increase the supply of drivers have quite powerful effects.

Ed Fountain, a veteran Uber driver in the Chicago area, said of the company and its messages to drivers, “It was always, constantly, trying to get you into a certain direction.” Brittany Jensen for The New York Times

Consider an algorithm called forward dispatch — Lyft has a similar one — that dispatches a new ride to a driver before the current one ends. Forward dispatch shortens waiting times for passengers, who may no longer have to wait for a driver 10 minutes away when a second driver is dropping off a passenger two minutes away.

Perhaps no less important, forward dispatch causes drivers to stay on the road substantially longer during busy periods — a key goal for both companies.

Uber and Lyft explain this in essentially the same way. “Drivers keep telling us the worst thing is when they’re idle for a long time,” said Kevin Fan, the director of product at Lyft. “If it’s slow, they’re going to go sign off. We want to make sure they’re constantly busy.”

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While this is unquestionably true, there is another way to think of the logic of forward dispatch: It overrides self-control.

Perhaps the most prominent example is that such automatic queuing appears to have fostered the rise of binge-watching on Netflix. “When one program is nearing the end of its running time, Netflix will automatically cue up the next episode in that series for you,” wrote the scholars Matthew Pittman and Kim Sheehan in a 2015 study of the phenomenon. “It requires very little effort to binge on Netflix; in fact, it takes more effort to stop than to keep going.”

As with viewers and binge-watching, many drivers appear to enjoy the forward-dispatch feature, which can increase earnings by keeping them busier. But it can also work against their interests by increasing the number of drivers on the road and defusing fare surges. And whether they enjoy it is separate from the question of agency — whether they have it, or whether the company does.

Uber officials say the feature initially produced so many rides at times that drivers began to experience a chronic Netflix ailment — the inability to stop for a bathroom break. Amid the uproar, Uber introduced a pause button.

“Drivers were saying: ‘I can never go offline. I’m on just continuous trips. This is a problem.’ So we redesigned it,” said Maya Choksi, a senior Uber official in charge of building products that help drivers. “In the middle of the trip, you can say, ‘Stop giving me requests.’ So you can have more control over when you want to stop driving.”

It is true that drivers can pause the services’ automatic queuing feature if they need to refill their tanks, or empty them, as the case may be. Yet once they log back in and accept their next ride, the feature kicks in again. To disable it, they would have to pause it every time they picked up a new passenger. By contrast, even Netflix allows users to permanently turn off its automatic queuing feature, known as Post-Play.

This pre-emptive hard-wiring can have a huge influence on behavior, said David Laibson, the chairman of the economics department at Harvard and a leading behavioral economist. Perhaps most notably, as Ms. Rosenblat and Luke Stark observed in an influential paper on these practices, Uber’s app does not let drivers see where a passenger is going before accepting the ride, making it hard to judge how profitable a trip will be.

Sometimes all that is necessary is the mere setting of a so-called default. Because humans tend to be governed by inertia, automatically enrolling them in retirement savings plans and then allowing them to opt out results in far higher participation than letting them opt in. Making Post-Play the default can have the same effect.

“If done right, these things can be socially beneficial,” Mr. Laibson said. “But you can think of all sorts of choice architecture that are quite contrary to human well-being.”

Even Mr. Hall, the Uber research director who downplayed the importance of behavioral economics to the company, did make at least one concession. “The optimal default we set is that we want you to do as much work as there is to do,” he said of the company’s app. “You’re not required to by any means. But that’s the default.”

Having more drivers on the road benefits ride-share companies, but drivers profit from surge pricing and scarcity in their ranks.

Ride-share companies, which do not bear the direct costs of drivers being idle, want to have as many drivers available as possible.

SIMULATION DETAILS: There are **10** drivers. During surge, prices increase by **2**, and passenger requests are reduced by **20%**. Passengers cancel after **20** minutes of waiting. Drivers cancel after **10** minutes. Total request: Regular fares cost **$1.00** a minute, driver expenses are **$0.65** a minute, and the app company takes **$0.50** leaving for you **20%** of the per-minute fare.

Imagining the Future

There are aspects of the platforms that genuinely do increase drivers’ control over their work lives, as Uber frequently points out. Unlike most workers, an Uber driver can put in a few hours each day between dropping children off at school and picking them up in the afternoon.

Uber is even in the process of developing a feature that allows drivers to tell the app in advance that they need to arrive at a given location at a given time. “If you need to pick up your kids at soccer practice at 6 p.m.,” said Nando Janakiram, the Uber official in charge of products that improve drivers’ experiences, “it will start to give you trips to take you in the general direction to get to a specific place in time.”

There is also the possibility that as the online gig economy matures, companies like Uber may adopt a set of norms that limit their ability to manipulate workers through cleverly designed apps.

Kelly Peters, chief executive of BEworks, a management consulting firm specializing in behavioral science, argued that the same data that makes it easier for Uber to nudge drivers into working an additional 30 or 60 minutes also makes it hard to escape the obligation to look after them.

For example, the company has access to a variety of metrics, like braking and acceleration speed, that indicate whether someone is driving erratically and may need to rest. “The next step may be individualized targeting and nudging in the moment,” Ms. Peters said. “‘Hey, you just got three passengers in a row who said they felt unsafe. Go home.’” Uber has already rolled out efforts in this vein in numerous cities.

That moment of maturity does not appear to have arrived yet, however. Consider a prompt that Uber rolled out this year, inviting drivers to press a large box if they want the app to navigate them to an area where they have a “higher chance” of finding passengers. The accompanying graphic resembles the one that indicates that an area’s fares are “surging,” except in this case fares are not necessarily higher.

Some drivers believe that the intent is to trick them into driving where Uber wants them to go, rather than where driving would be most profitable, by implying that they will find a surge there. “They’re trying to move people where they want them,” said Mr. Weber, the Tampa-area driver. “But you get there and it’s nothing. It happens all the time.” Mr. Weber noted that the design of the graphic makes the prompt much easier to accept than decline, which requires pressing a small rectangle in the top left corner.

Uber said that the feature was an experiment intended primarily to help new drivers who frequently say they do not know where to find passengers, and that it could be changed if drivers were dissatisfied.

Individual features aside, the broader question of how much Uber seeks to influence drivers through behavioral science may come down to how much its business model requires it.

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While the company has made no secret of its investment in self-driving cars, it could be a decade or more before they completely replace human drivers. In the meantime, as long as Uber continues to set growth and passenger volume as critical goals, it will have an incentive to make wringing more hours out of drivers a higher priority than the drivers’ bottom line whenever it faces a close call between the two.

It will also have an incentive to obtain these hours as cheaply as possible. And there is simply no cheaper way than hiring contractors and nudging them to drive when and where they are needed. Industry insiders estimate that relying on independent contractors rather than employees can lower direct costs by roughly 25 percent.

Moreover, the contractor model itself provides a strong impetus for companies like Uber to grow. Many companies in the gig economy simply do not have enough workers, or rich enough data about their workers’ behavior, to navigate busy periods using nudges and the like. To avoid chronic understaffing, they have switched to an employee model that allows them to compel workers to log in when the companies most need them.

Once companies achieve a certain scale, on the other hand, they enter a virtuous cycle: The risk of understaffing drops with a big enough pool of workers, and the cost savings of using contractors begins to outweigh the inefficiencies. This in turn frees up money to enter new markets and acquire new customers, which makes the contractor model still more efficient, and throws off still more savings.

It is, as a result, not too hard to imagine a future in which massive digital platforms like Uber have an appetite for tens of millions of workers — not only for ferrying people, but also for delivering food and retail goods. Nor is it hard to imagine workers’ obliging them, perhaps because their skills do not match the needs of traditional employers, or because they need to supplement their wages.

In such an economy, experts say, using big data and algorithms to manage workers will not simply be a niche phenomenon. It may become one of the most common ways of managing the American labor force.

“You have all these players entering into this space, and the assumption is they’ll do it through vast armies of underemployed people looking for extra hours, and we can control every nuance about what they do but not have to pay them,” said David Weil, the top wage-and-hour official under President Barack Obama.

When you stop to consider the enormous cost advantages, Mr. Weil said, “it says to me this is an area that will grow fast.”

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